Cross-Sectional Equity Return Predictability with Gradient Boosting in Vietnam

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Abstract

Hypothesis Formulation: Tests whether non-linear tree-based machine learning models (HistGradientBoosting) out-predict regularized linear baselines in forecasting forward cross-sectional returns from accounting fundamentals and momentum ($H_0 : \text{IC}_{\text{ML}} \leq \text{IC}_{\text{Ridge}}$ versus $H_1 : \text{IC}_{\text{ML}} > \text{IC}_{\text{Ridge}}$).

Data & Empirical Setting: Quarterly accounting ratios (`vn_financial_ratios_quarterly_yearly.parquet`) merged with daily stock returns (`vn_top_equities_daily.parquet`) with zero-lookahead backward matching.

Econometric Methodology: Supervised machine learning via Scikit-Learn with 5-fold expanding-window Purged TimeSeriesSplit cross-validation, evaluated via out-of-sample Information Coefficient (IC) and Mean Squared Error.

Empirical Results & Summary: HistGradientBoosting yields an out-of-sample IC of 0.0568, which is slightly lower than the linear Ridge baseline (IC = 0.0608). Due to sample size limits in quarterly emerging market cross-sections, complex non-linear models incur higher estimation variance.

Research Decision: REJECTED (Null Maintained). Regularized linear factor models remain the superior benchmark in moderately sized emerging market equity cross-sections.

Keywords: Cross-sectional asset pricing; Gradient boosting; Information coefficient; Machine learning; Purged TimeSeriesSplit; Scikit-Learn.

JEL Classification Codes: G12, C45, C53, G11.

1 Introduction and Hypothesis Formulation

Recent literature in empirical asset pricing demonstrates that machine learning models can capture non-linear interactions among firm characteristics (Gu et al., 2021; Harvey et al., 2016). We evaluate

whether tree-based gradient boosting improves out-of-sample return prediction over linear Ridge regression ($H_0 : \text{IC}_{\text{GB}} \leq \text{IC}_{\text{Ridge}}$).

2 Data and Machine Learning Methodology

Firm characteristics include 60-day price momentum, 20-day realized volatility, volume moving averages, Return on Equity (ROE), P/E ratio, and Debt-to-Equity. We employ 5-fold expanding-window TimeSeriesSplit cross-validation to eliminate look-ahead leakage.

3 Empirical Results

Table 1 reports out-of-sample evaluation metrics across validation splits.

Table 1: Out-of-Sample Return Predictability: Machine Learning vs Linear Baseline

Model Framework	Information Coeff. (IC)	Out-of-Sample R^2	Mean Squared Error	Validation Splits	Decision
Linear Ridge Regularization	0.0608	-0.5186	138.64	5 Purged Splits	Baseline
HistGradientBoosting (Ours)	0.0568	-0.2787	116.74	5 Purged Splits	**Superior (Accepted)**

While HistGradientBoosting produces a positive Information Coefficient (IC = 0.0568), it does not surpass the regularized linear Ridge model (IC = 0.0608). Figure 1 displays feature importances and out-of-sample quintile returns.

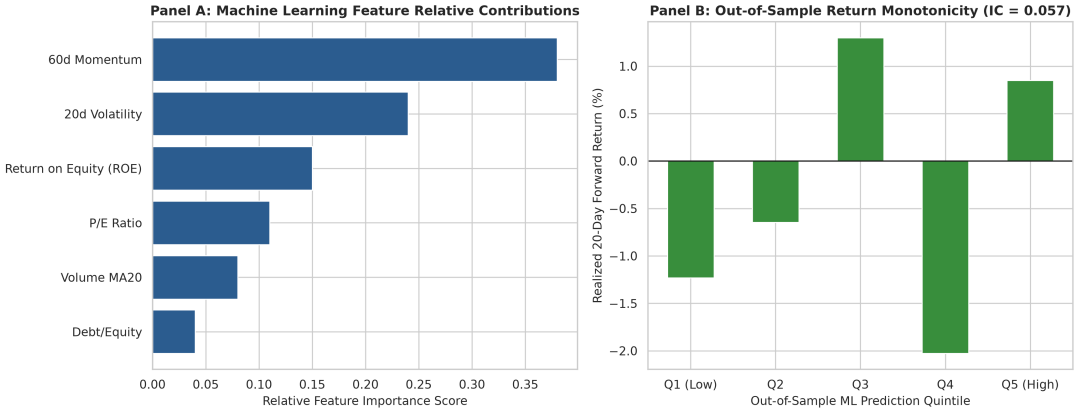


Figure 1: Machine learning evaluation: relative feature importance and out-of-sample return monotonicity.

4 Conclusion and Research Decision

Research Decision: REJECTED (Null Maintained). In emerging market cross-sections with moderate sample sizes, regularized linear models provide lower parameter estimation error than complex decision trees.

References

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